Received: 16 August 2017 | Revised: 22 December 2017 | Accepted: 28 December 2017

DOI: 10.1111/jsr.12667

REGULAR RESEARCH PAPER

# Development and validation of an algorithm for the study of sleep using a biometric shirt in young healthy adults

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**Funding information**
Fonds de Recherche du Québec - Nature et Technologies; Natural Sciences and Engineering Research Council of Canada, Grant/Award Number: 1258630; Fondation Les petits trésors de l'Hôpital Rivière-des-Prairies

## Summary

Portable polysomnography is often too complex and encumbering for recording sleep at home. We recorded sleep using a biometric shirt (electrocardiogram sensors, respiratory inductance plethysmography bands and an accelerometer) in 21 healthy young adults recorded in a sleep laboratory for two consecutive nights, together with standard polysomnography. Polysomnographic recordings were scored using standard methods. An algorithm was developed to classify the biometric shirt recordings into rapid eye movement sleep, non-rapid eye movement sleep and wake. The algorithm was based on breathing rate and heart rate variability, body movement, and included a correction for sleep onset and offset. The overall mean percentage of agreement between the two sets of recordings was 77.4%; when non-rapid eye movement and rapid eye movement sleep epochs were grouped together, it increased to 90.8%. The overall kappa coefficient was 0.53. Five of the seven sleep variables were significantly correlated. The findings of this pilot study indicate that this simple portable system could be used to estimate the general sleep pattern of young healthy adults.

KEYWORDS
actigraphy, home recording, telemetry, wearable technology

## 1 | INTRODUCTION

The gold-standard for assessing sleep is polysomnography (PSG), thanks to its capability to record the numerous physiological signals required to classify sleep stages, evaluate sleep quality and diagnose sleep disorders. Significant efforts have been invested over the last decade to simplify and automate the scoring and interpretation of PSG data (Krakovská & Mezeiová, 2011). Portable PSG for in-home recordings has been proposed, but the technology is still complex, encumbering and requires the intervention of skilled technologists (Portier et al., 2000).

A simpler approach is to use only one PSG signal. Adnane, Jiang, and Yan (2012) used electrocardiography (ECG) to perform a sleep/awake classification with a learning system, and obtained a moderate agreement with standard PSG. Kesper, Canisius, Penzel, Ploch, and Cassel (2012) identified wake, rapid eye movement (REM) sleep, light sleep and deep sleep using three heart rate (HR) parameters, and obtained a 57.8% agreement with standard PSG. A combination of sensor belts for respiration and HR was proposed (Choi & Jiang, 2008) as well as ballistocardiography systems measuring HR, breathing rate (BR) and musculoskeletal movements (Mendez, Matteucci, Cerutti, Bianchi, & Kortelainen, 2009), but the quality of the recorded signal or the accuracy of sleep states identification still need to be validated. Pneumatic recording techniques measuring heartbeats and movement signals have been shown to be able to classify three-six sleep stages with low to moderate agreement with standard PSG (Kurihara & Watanabe, 2012). Headbands recording electroencephalography (EEG), electro-oculography (EOG) and electromyography (EMG) can identify wake, REM, light and deep sleep with a good agreement (~60%) against standard PSG (Shambroom, FáBregas, & Johnstone, 2012), but this system is no longer manufactured.

The purpose of the present study was to evaluate the feasibility of using ECG, respiration and body movement signals from a


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biometric shirt to identify three vigilance states (wake, non-rapid eye movement [NREM] sleep, REM sleep) and quantify sleep efficiency. This biometric shirt has been validated for sport training and day-to-day usage (Villar, Beltrame, & Hughson, 2015). We hypothesized that the classification performance of vigilance states using this technology would approach that of standard PSG.

## 2 | MATERIALS AND METHODS

### 2.1 | Participants

Twenty healthy young adults (12 women, eight men, 21.9 ± 2.0 years old, range from 19 to 25 years) were recruited via public advertisement. Exclusion criteria were a past or current history of psychiatric, neurological or other medical condition, the use of central nervous system (CNS)-active medications, reporting the presence of a sleep disorder, having experienced shift work, night work or travelling outside the local time zone for the last month. Participants provided written informed consent and were compensated financially for their participation in the study. The protocol was approved by the Research Ethics Review Boards of the Hôpital Rivière-des-Prairies and of Polytechnique Montréal (project 13-11P).

### 2.2 | Sleep recordings

Participants were required to remain abstinent from alcohol, drugs and caffeine for 12 hr before each recording. They were recorded in individual bedrooms for two consecutive nights in a sleep laboratory. Participants went to bed and were awakened according to their preferred time, and resumed their normal daily activities in the morning. The data from the 2 nights of recordings were used in the data analysis.

Polysomnography included EOG, submental EMG and frontal, central and occipital EEG, as well as anterior tibialis EMG, ECG (Lead I infraclavicular fossae), oro-nasal airflow with pressure transducers (Ptaf Lite, Pro-tech, Mukilteo, WA, USA), respiratory effort with thoracic and abdominal respiratory inductance plethysmography (RIP) belts (zRIP, Pro-tech Services, Mukilteo, WA, USA) and arterial oxygen saturation with infrared oxymetry sensors to the index finger (Datex Ohmeda 3900 Pulse Oxymeter, Datex-Ohmeda, Louisville, CO, USA). All recording sessions were accompanied by audiovisual monitoring.

Sleep was scored by certified and experienced sleep technologists according to Rechtschaffen and Kales (1968), using 20-s epochs. This yielded the identification of six vigilance states: Wake, stage 1 (NREM1), stage 2 (NREM2), stage 3 (NREM3), stage 4 (NREM4) and REM sleep. Sleep-onset latency was defined as the time from the start of the recording to the first occurrence of any sleep stage except NREM1. The sleep period time was defined as the duration between sleep onset and the final awakening. Total sleep time (TST) was the total number of minutes spent in any of the sleep stages during the sleep period. Sleep efficiency was equal to the percentage of sleep time during the sleep period.

### 2.3 | Biometric shirt

The biometric shirt system (BioSS) is shown in Figure 1. It is composed of a shirt with embedded sensors and a detachable device containing a microcontroller performing signal acquisition, data processing and communication protocols, and a three-axis accelerometer (Hexoskin, model HX1, Carré Technologies, Montréal, Canada). The shirt size was individually selected for each participant. Three silver-coated stretchable fabric sensors record an ECG lead that corresponded approximately to the V5 lead (Figure 1b). Two RIP bands integrated into the biometric shirt recorded abdominal and thoracic respiratory movements (Figure 1b). The sensor wires were inserted in a detachable device, connected to the shirt and placed into its pocket (Figure 1c,d). Connecting the device to the shirt automatically started a recording. The sampling rates for the ECG, respiration and accelerometer signals were 256 Hz, 128 Hz and 64 Hz, respectively.

Data recorded from the PSG equipment and from the biometric shirt were collected simultaneously for each participant. At the beginning and the end of each recording, a biocalibration provided baseline for each channel. Six consecutive high-amplitude respirations were performed to synchronize both recording systems.

### 2.4 | Biometric shirt signal preprocessing

Signal preprocessing was performed by the BioSS device to compute HR, BR and activity values throughout the recording. The device was connected to a computer after each recording to transfer the data to the BioSS server. After the initial detection of the QRS complexes on the ECG, false detections were rejected based on temporal criteria to obtain corrected RR intervals. BR was computed each second as the average of the last seven breathing intervals. After preprocessing, the raw data and the preprocessed data were downloaded to a personal computer for the development of the sleep analysis algorithm. Figure 2 shows an example of the raw data obtained with the BioSS as displayed on a dashboard. HR, BR and body movement data were computed in 20-s epochs in order to match scoring epochs of PSG data. For each 20-s epoch, HR corresponded to the average value of the RR intervals in the epochs. When an epoch did not include a valid RR interval due to movement artefacts, the last valid interval value was associated to this epoch. BR and body movement values were computed as the mean of the values obtained during each second. Nights that included two or more 5-min segments of artefacted data were excluded.

### 2.5 | Classification algorithm

The classification algorithm used the three parameters described above to discriminate between three vigilance states: wake, NREM sleep and REM sleep. This algorithm was partly derived from the method of Kurihara and Watanabe (2012) who used HR and body movement signals recorded from a pneumatic sensor placed under a mattress. The premise of this method is that higher HR is associated to REM sleep, and that body movements become smaller and less

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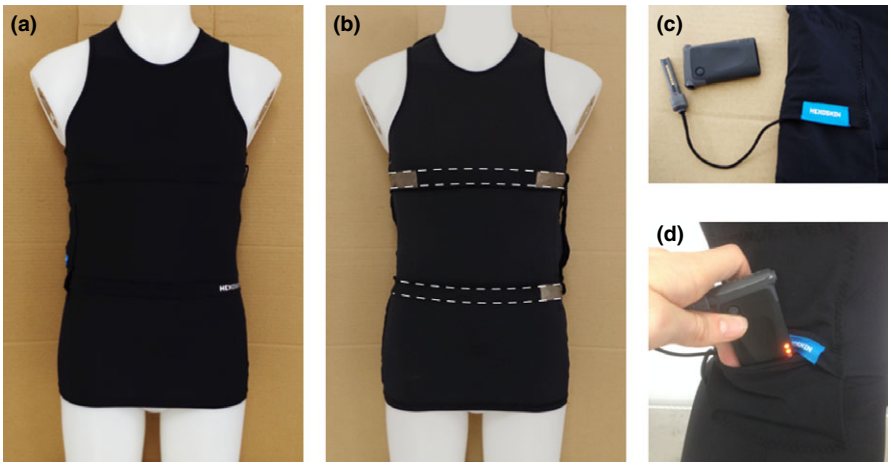


**FIGURE 1** Biometric shirt system. (a) Shirt front view. (b) Shirt inside out showing respiration bands and electrocardiogram (ECG). (c) Connector and device. (d) Device pocket

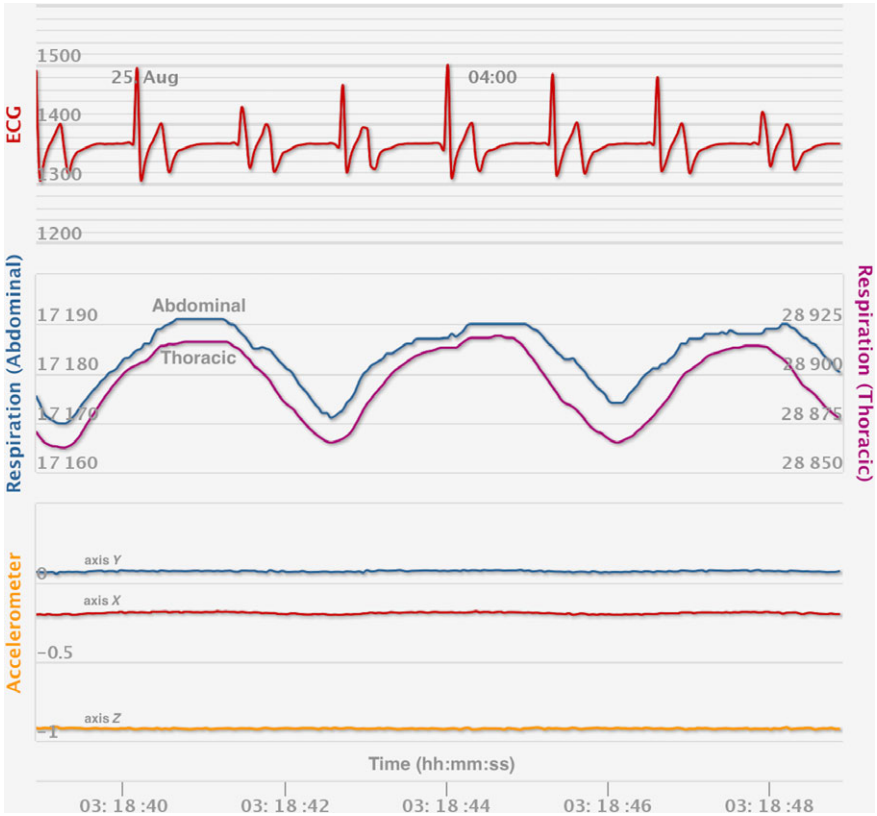


**FIGURE 2** Dashboard visualization of the raw data

frequent when sleep deepens from wake to NREM sleep. HR and body movement signals obtained with the biometric shirt during the present study were processed using the same premises. As accelerometer signals are almost constant during sleep (with changes occurring mostly during posture shifts), the body movement variable was computed with a different formula (see below). Moreover, the biometric shirt provided respiration signals that were added into the algorithm. Thus, our algorithm was based on four hypotheses: (1) HR variability (HRV) is higher during REM sleep compared with NREM (Choi & Jiang, 2008); (2) HR is higher with higher HRV during the Wake stage (Dumont et al., 2004); (3) BR has the same tendency as the HR as HR is influenced by respiration (Chung et al., 2009); (4) less movements occur during the deepest NREM stages, and more movements are associated with the Wake state (Kurihara & Watanabe, 2012).

The classification parameters *Rh*(*i*), *Rb*(*i*) and *D*(*i*) were fitted into 20-s epochs. *Rh*(*i*) was computed using the following equation to measure HRV:

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$$Rh(i) = \frac{1}{2q+1}\sum_{j=-q}^{q}|\mathrm{HR}_{i-1+j} - \mathrm{HR}_{i+1+j}| \quad (1)$$

where for each epoch $i$, the absolute value of the difference between the average HR of the following epoch and that of the preceding epoch is filtered using a moving average filter with an index $j$ and a range parameter $q = 20$. $Rb(i)$ was computed using a similar equation to measure BR variability:

$$Rb(i) = \frac{1}{2q+1}\sum_{j=-q}^{q}|\mathrm{BR}_{i-1+j} - \mathrm{BR}_{i+1+j}| \quad (2)$$

$D(i)$ measured body movement. A logarithmic scale was used to accentuate the smaller values, and a constant $p$ (set to 0.01) was added to the mean body movement amplitude during the epoch ($A_i^{\mathrm{activity}}$), to eliminate the singularity for a null body movement value:

$$D(i) = \log_2(p + A_i^{\mathrm{activity}}) \quad (3)$$

Figure 3 illustrates how these three parameters vary across the night and how they are related to PSG-defined sleep. The corresponding HR, BR and body movement data are also presented.

Figure 3c shows the decision tree of the basic sleep stage classification algorithm. For each epoch $i$, the whole three tests applied to the three parameters must be met to classify the epoch as REM sleep. If one condition is not met, a second test is applied to the $D(i)$ parameter to distinguish Wake from NREM. Algorithms excluding either the $Rb(i)$ parameter or the $Rh(i)$ parameter from the first test were performed to assess the respective contributions of the cardiac and respiratory data.

For each parameter, a threshold was defined as the mean value of that parameter computed during an entire recording multiplied by a coefficient that was the same for all participants. To prevent overfitting, the $k1$ and $k2$ coefficients were optimized independently by maximizing the kappa value in a leave-one-out cross-validation. The $k3$ coefficient was pre-determined by observing the $D$ data and the same value was used in all the algorithms. The $k1$ and $k2$ coefficients were optimized independently, for all the algorithms tested. Table S1 shows the values of the $k$ coefficients for the different algorithms.

During the optimization process, segregation issues between Wake and REM stages at the beginning and at the end of the night were observed. To address this problem, two corrections were added to the algorithm. For the sleep-onset correction, the hypothesis was that the sleep period would start in NREM sleep, and not in REM sleep. This is generally true for healthy sleepers with normal sleeping habits (Roebuck et al., 2014). In accordance with Kesper et al. (2012), the first 4 min of a recording were always assigned to the Wake stage and the beginning of the first 4 consecutive minutes in NREM stage was associated with sleep onset. All preceding epochs were classified as Wake. Identifying the end of the night was more complex because the final awakening could occur either in REM or NREM sleep. By default, the last segment of sleep was entered as the sleep stage lasting at least 2 consecutive minutes, and the final last 4 min of a recording were always assigned to the Wake stage.

### 2.6 | Classification agreement and statistical analysis

The six vigilance states identified using the PSG recordings were grouped into three sleep stages. All epochs scored as NREM1 before any other sleep stage occurred were classified as Wake as sleep onset was defined as the occurrence of any sleep stage other than NREM1. The four NREM sleep stages were combined into a single NREM category. Some epochs at the beginning and end of the PSG or BioSS recordings were discarded to ensure a perfect temporal alignment and equal duration of both sets of data.

The agreement between the BioSS and PSG classifications was assessed epoch by epoch. Contingency tables were constructed for each algorithm tested, where the numbers of epochs scored similarly and differently by the two methods were counted. The total percentage of agreement was the ratio of all epochs classified similarly by the two methods divided by the total number of epochs × 100. The percentage agreement for each sleep stage was the ratio of all epochs classified in this sleep stage by the two methods divided by the total number of epochs classified in this sleep stage by the PSG × 100. The kappa coefficient was also used because percentages of agreement do not take into account the proportion of each vigilance state (Tilmanne, Urbain, Kothare, Vande Wouwer, & Kothare, 2009). A kappa value of 0 is considered as no agreement, [0–0.2] as a slight agreement, [0.2–0.4] as a fair agreement, [0.4–0.6] as a moderate agreement, [0.6–0.8] as a substantial agreement, and [0.8–1.0] as an almost perfect agreement (McGinn et al., 2004).

Seven sleep variables were computed: sleep latency, TST, the time spent in each sleep stage, the sleep period, the wake time after sleep onset, and the sleep efficiency. The values obtained with the classifications methods (BioSS and PSG) were compared using a paired Student's $t$ test.

Two statistical tests were applied to the kappa coefficient. Two-factor within-subjects ANOVAs for repeated measures were used to evaluate the similarity between the six algorithms listed in Table S1. The dependent variable was the kappa value, the first grouping variable was the participant's number allowing paired comparisons, and the two factors were the correction and the classification algorithm. A Student's $t$ test ($p < .05$) assessed the relative contributions of HR and BR to the performance of the classification algorithms. Statistic tests and corresponding graphics were made with Matlab (*The MathWorks R2010a*) using default parameters.

## 3 | RESULTS

Five nights from four participants were rejected due to recurrent signal artefacts caused by an ill-fitting or misplaced shirt. A typical good-quality bioshirt recording is shown in Figure 3. The HR and BR

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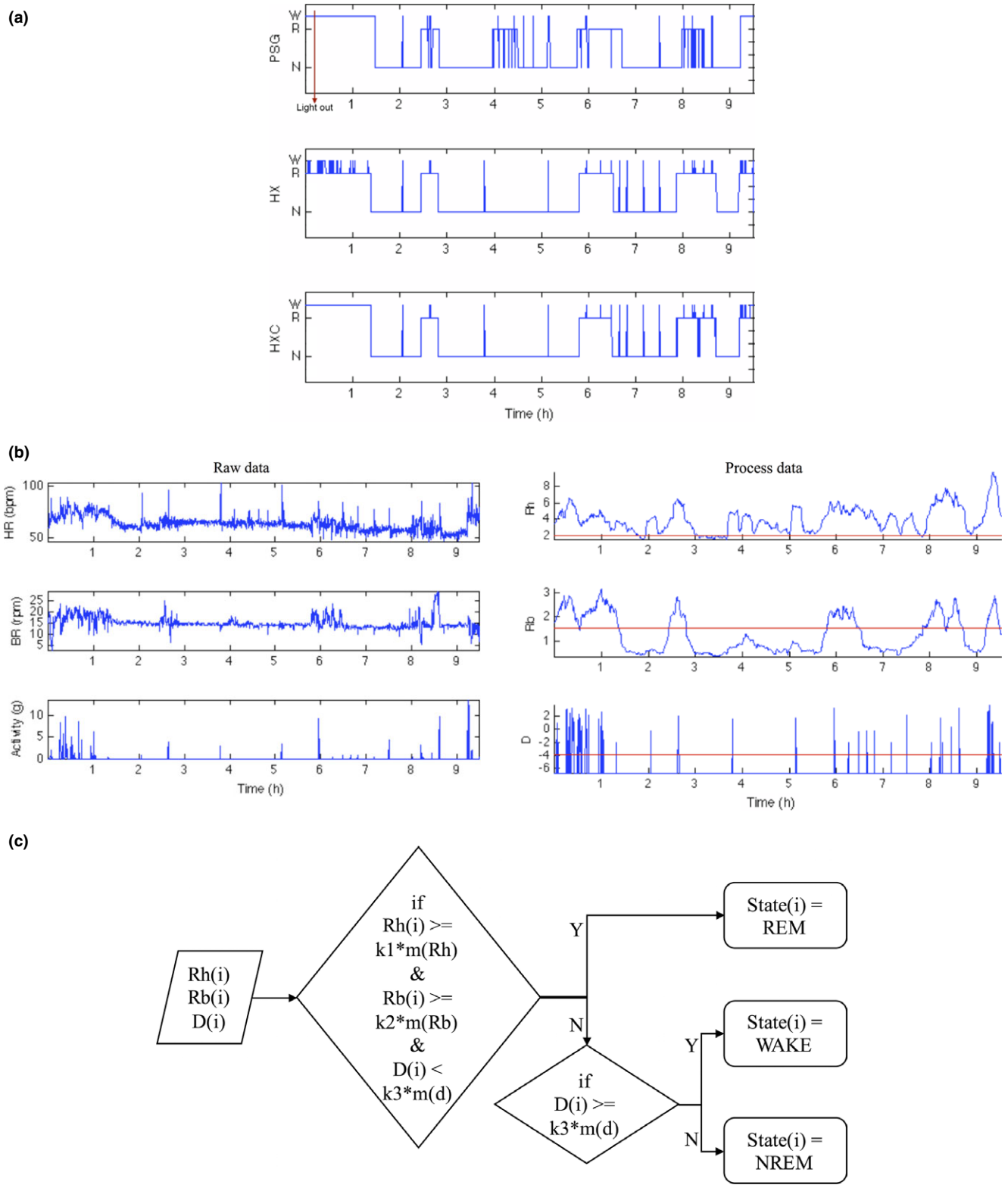


**FIGURE 3** (a) Typical examples of vigilance states as determined by three classification systems: PSG = classification according to Rechtschaffen and Kales (1968); HX = basic BioSS classification algorithm; HXC = BioSS classification algorithm with onset and offset correction. W, wake; R, REM sleep; N, NREM sleep. (b) Classification parameters and processed data from a 23-year-old female subject. For the raw data, the horizontal red lines correspond to the classification threshold values. *Rh* = HR variability; *Rb* = BR variability; *D* = body movements. The filtered, processed values are the mean value for each 20-s epoch from the raw data. (c) Decision tree for the classification algorithm of vigilance states. For each 20-s epoch *i*, *Rh*(*i*) reflects HR variability, *Rb*(*i*) reflects BR variability, *D*(*i*) reflects body movement. *k*1, *k*2 and *k*3 threshold coefficients each multiply the mean of their respective parameter as computed during the entire recording: m(*Rh*), m(*Rb*) and m(*D*), respectively. NREM, non-rapid eye movement; PSG, polysomnogram; REM, rapid eye movement

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**TABLE 1** Contingency tables for epoch-by-epoch classification results between PSG and each of the two algorithms

| | | PSG | | | | | PSG | | |
|---|---|---|---|---|---|---|---|---|---|
| | | W | R | N | | | W | R | N |
| HX | W | 2,079 | 310 | 1,458 | HXC | W | 5,689 | 313 | 1,692 |
| | R | 4,852 | 5,951 | 5,242 | | R | 1,716 | 5,622 | 4,404 |
| | N | 2,172 | 3,101 | 33,433 | | N | 1,698 | 3,427 | 34,037 |
| | Total = 58,598 epochs | | | | | Total = 58,598 epochs | | | |

PSG, polysomnography; W, Wake; R, REM sleep; N, non-REM sleep; HX, basic algorithm; HXC, HX with onset and offset correction. NREM, non-rapid eye movement; REM, rapid eye movement.

curves show the typical higher values with larger variations at both extremities of the recording. Figure 3 also presents three hypnograms corresponding to the same recording, with vigilance states classification according to Rechtschaffen and Kales (1968), the basic BioSS classification algorithm, and the BioSS classification algorithm with onset and offset correction, respectively. The general sleep pattern obtained with the basic BioSS algorithm (HX) is similar to that obtained with the classical PSG scoring criteria, except that the wake periods during sleep latency and during the last awakening are intermingled with faulty REM sleep epochs; the second REM period is also undetected. The addition of corrections for the sleep-onset and -offset periods (HXC) perfectly improved the detection of wake during the sleep latency period, but the sleep-offset detection still needs improvement. This correction did not address the second REM sleep period that is still missing. The segments with slow and regular HR and BR are clearly associated with NREM sleep, whereas segments with faster and irregular rates are associated with Wake and REM sleep (Figure 3).

## 3.1 | Agreement between sleep classification systems

Table 1 shows the contingency tables for all the epochs of the 35 recordings, for a total of 58,598 epochs. The basic BioSS algorithm without the corrections for sleep onset and offset generated a total agreement of 70.2%, while adding the sleep-onset and -offset corrections increased it to 77.4%. When NREM and REM sleep epochs were grouped together, the total agreements without and with corrections were 85.0% and 90.8%, respectively.

Figure 4 illustrates the detailed statistical results obtained with the HXC algorithm with corrections for sleep onset and offset for the 35 recordings using box-plot diagrams. The average (±SD) agreements were 85.1 ± 5.9% for NREM sleep, 59.1 ± 21.4% for REM sleep, 63.0 ± 13.4% for Wake, and 95.5 ± 2.2% for the combination of REM and NREM sleep. The average (±SD) kappa was 0.53 ± 0.13, which is considered as a moderate agreement between two classifications systems (see also the Bland–Altman plot in Figure S1).

To evaluate the respective contributions of the cardiac and respiratory data, four more algorithms were tested: BR and movement (HXRb); HR and movement (HXRh); and each of these two

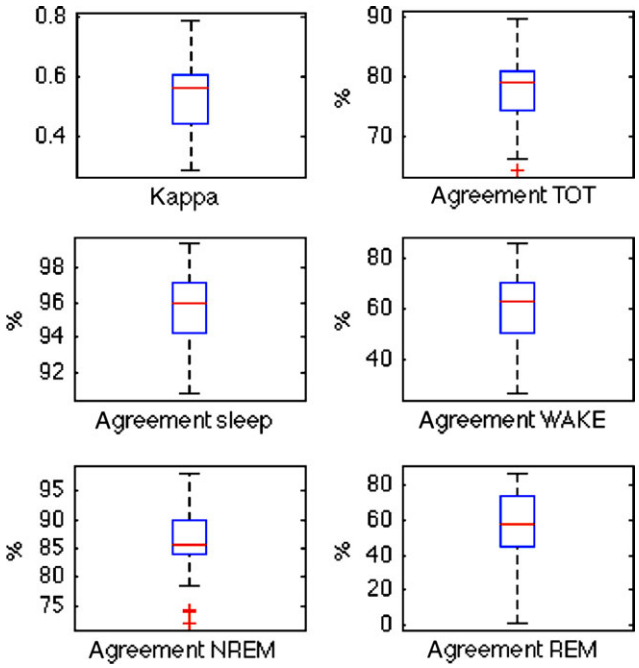


**FIGURE 4** Box-plot diagram of kappa coefficient and % agreements. The kappa coefficient and % agreement between the HXC algorithm and the polysomnogram (PSG) classifications, computed for the 35 recordings. Box edges = inter-quartile ranges; box horizontal bars = median; whiskers = extremes values

algorithms with an additional correction for the sleep-onset and sleep-offset periods (HXRbC, HXRhC). Kappa values for these six algorithms are presented online (Figure S2). The ANOVA test disclosed significant differences between the two factors, i.e. the correction factor and the classification algorithm factor ($F = 17.64$, $p = 6.76^{E-7}$). The use of onset and offset correction increased significantly the classification agreement with the PSG (HX: $K = 0.40 \pm 0.12$ and HXC: $K = 0.53 \pm 0.13$; $t = -13.78$, $p = 1.77^{E-15}$). Student's $t$ test showed that the elimination of the BR data from the basic algorithm decreased significantly the classification agreement as measured by the kappa coefficient (HXRhC: $K = 0.34 \pm 0.13$; $t = 9.40$, $p = 5.54^{E-11}$), whereas the elimination of the HR data from the basic algorithm did not significantly modify the classification agreement (HXRbC: $K = 0.54 \pm 0.13$; $t = -1.53$, $p = .1358$).

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### 3.2 | Correlation between sleep classification systems

Seven sleep measures were tested: sleep latency, TST, the time spent in each sleep stage, the sleep period, the wake time after sleep onset, and the sleep efficiency. Table 2 presents the metrics means and standard deviations for all the recordings for the PSG and the HX algorithm without and with corrections. The measured metrics that are significantly different ($p < .05$) than the PSG results are identified with an asterisk.

To evaluate the correlation between the sleep metrics from the HXC corrected algorithm and PSG, scatter-plots for seven sleep metrics were also plotted, and the Pearson's correlation coefficients ($r$) were calculated for the 35 recordings (Figure 5). Five of the seven metrics were significantly correlated ($p < .01$), whereas two were not correlated (Time in NREM, and Time in REM).

## 4 | DISCUSSION

The visual analysis of the hypnograms illustrates that the HXC algorithm achieved a good general agreement with the standard PSG method (Rechtschaffen & Kales, 1968). The best algorithm (HXC) was based on BR variability (*Rb*), HRV (*Rh*), and body movements with corrections for sleep onset and offset. The mean general agreement between the HXC algorithm and PSG was high, and the kappa values ranged from fair to substantial. As a matter of fact, the data from five of the seven sleep variables generated by PSG and HXC methods were significantly correlated, with the highest values obtained on sleep latency and sleep period. The agreement between the HXC and PSG classifications varied according to sleep stages. The highest values were found with NREM sleep and the lowest values were found with REM sleep. When NREM and REM sleep were grouped together, the agreement between PSG and the biometric shirt results reached even higher levels.

**TABLE 2** Sleep metrics from the different classification methods

| Mean ± SD | PSG | HX | HXC |
|---|---|---|---|
| Sleep latency (min) | 42.8 ± 21.6 | 0.69* ± 0.55 | 39.1 ± 22.7 |
| TST (min) | 471.4 ± 29.0 | 521.4* ± 23.5 | 485.9* ± 25.3 |
| Time in REM sleep (min) | 89.2 ± 25.4 | 162.4* ± 30.7 | 105.4* ± 25.0 |
| Time in NREM sleep (min) | 382.2 ± 34.7 | 359.1* ± 36.6 | 380.5 ± 31.7 |
| Total sleep period (min) | 496.9 ± 29.7 | 557.6* ± 27.1 | 511.6* ± 28.9 |
| Wake after sleep onset (min) | 25.5 ± 22.3 | 36.1* ± 12.5 | 25.7 ± 9.8 |
| Sleep efficiency (%) | 95.0 ± 4.3 | 93.6* ± 2.1 | 95.0 ± 1.8 |

PSG, polysomnography; HX, basic algorithm; HXC, HX with onset and offset correction. NREM, non-rapid eye movement; REM, rapid eye movement; TST, total sleep time.
*Significantly different from PSG results ($p < .05$).

The data suggest that BR offered a better contribution than HR to the classification accuracy based on the biometric shirt data. Removing BR data from the initial algorithm significantly decreased the classification agreement as measured by the kappa coefficient, whereas removing the HR data did not. Breathing patterns have been previously shown to be a powerful index to classify sleep stages because sympathetic activation during REM sleep typically results in irregular and rapid breathing patterns (Chung et al., 2009), and the parasympathetic shift upon NREM results in a deeper and more regular pattern (Cabiddu, Cerutti, Viardot, Werner, & Bianchi, 2012). Moreover, a long-range correlation of the breath-to-breath intervals is observed during REM sleep and wake state, while it is absent during NREM sleep (Bartsch, Kantelhardt, Penzel, & Havlin, 2007). This behaviour is also true for the cardiac beat-to-beat intervals, as correlations between sympathetic and parasympathetic nervous system activities and sleep stages have also been reported using HRV as an index of sympatho-vagal balance (Cabiddu et al., 2012; Choi & Jiang, 2008). The phase synchronization between heartbeat and breathing is higher during NREM sleep and reduced during REM sleep (Bartsch et al., 2007; Cabiddu et al., 2012). Bartsch et al. (2007) thus concluded that the physiological synchronization mechanism of the cardiorespiratory system is mainly initiated by the influence of the respiratory cycle on HR and not the other way around.

Challenges remain, however. Somewhat like actigraphy, long wake periods during the night proved to be difficult classifying, possibly due to immobility. Because biometric shirts generate cardiorespiratory data in addition to accelerometry, future developments should optimize the use of such data to improve the detection of long wake periods. The algorithms tested here had difficulties detecting Wake epochs at the beginning and at the end of the night, being sometimes misclassified as REM sleep periods possibly because cardiovascular activity is comparable in these two vigilance states (Roebuck et al., 2014). The sleep-onset and -offset correction addressed this challenge and significantly improved the results. Still, the lowest agreement results between PSG and the biometric shirt occurred in cases of short REM sleep periods. This is possibly because the transition from NREM sleep to REM sleep is characterized by a fast increase of HR, while the transition from REM sleep to NREM sleep is characterized by a more gradual decrease (Dumont et al., 2004). Our data show indeed that the agreements between PSG and the biometric shirt were better when REM sleep distribution was closer to standard values. This is a good starting point to determine what should be improved next in trying to detect irregular patterns of sleep stages with a biometric shirt system as the one tested here.

Still, biometric shirts bear many advantages for sleep monitoring at home. The BioSS regroups true ECG, two RIP bands and an accelerometer with the sensors embedded in the shirt fabric itself. The system is portable – actually wearable, versatile, readily available through regular trade and the setup is user friendly. This is a resourceful tool for multipurpose perspectives.

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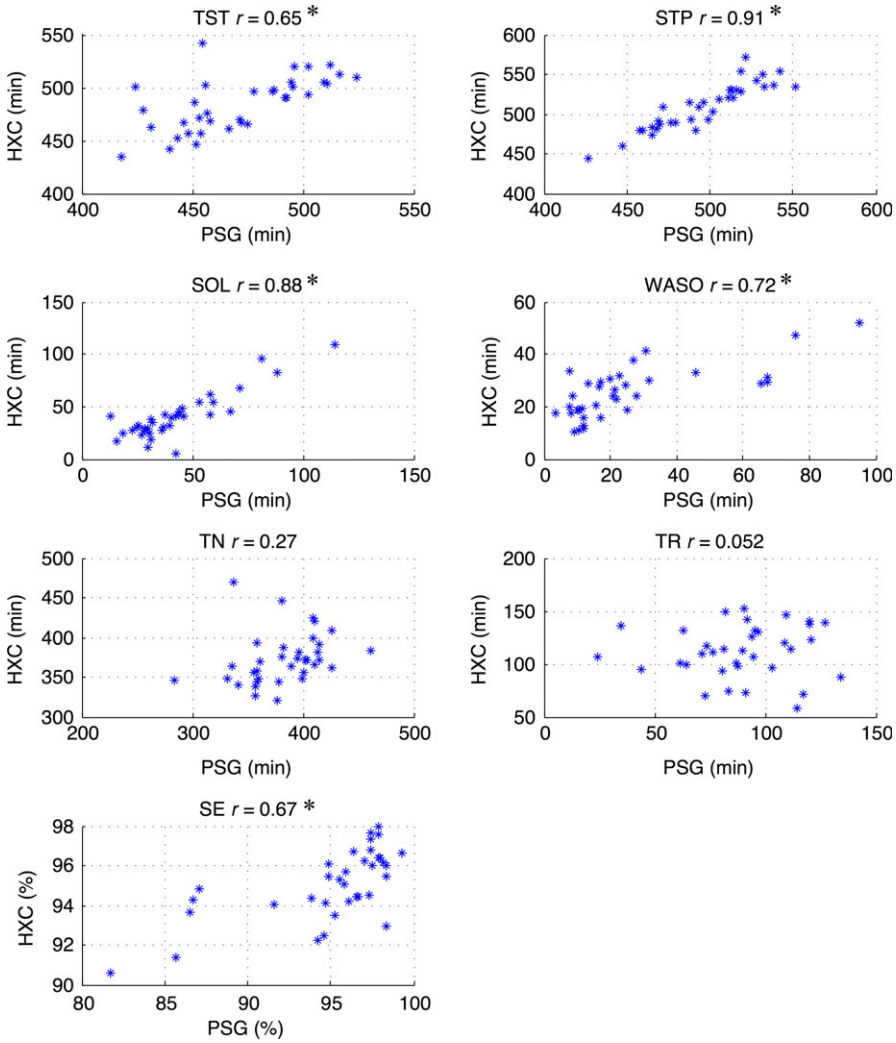


**FIGURE 5** Scatter-plots of the sleep metrics. Each point represents one of the 35 recordings using the HXC algorithm and the polysomnogram (PSG) classification. TST, total sleep time; STP, sleep total period; SL, sleep latency; WASO, wake after sleep onset; TR, time in REM; TN, time in NREM; SE, sleep efficiency. *r* = Pearson's correlation coefficient. * = significant correlations. NREM, non-rapid eye movement; REM, rapid eye movement

The biometric shirt system used here compares favourably with other recently published systems. Kurihara and Watanabe (2012) recorded HR and body movement using an air mattress placed under a bed cushion connected to a highly sensitive pressure sensor in order to classify Wake, NREM and REM sleep into 1-min epochs. The average agreement and kappa values compared with PSG (77.5% and 0.48, respectively) were quite similar to the values obtained here with the HXC algorithm (77.4% and 0.53, respectively). Such pneumatic recording techniques, however, are not truly ambulatory, and generate a single signal (mattress pressure changes) so that HR and movement data need to be sorted out and estimated using further signal processing.

The present study bears some limitations that should be considered. The participants were young, healthy sleepers so the results cannot be extrapolated to the general population and pathological conditions. The addition of sleep-onset and -offset corrections to the HXC algorithm prevents the application of this algorithm to sleep disorders involving rapid transitions between Wake and REM sleep. The lack of significant correlations between the biometric shirt and PSG in two out of seven sleep variables indicates that the new technology could be improved further.

In conclusion, the present study showed the feasibility of using a biometric shirt system to record physiological data during sleep and automatically classifying three vigilance states (wake, NREM sleep and REM sleep) with moderate to high agreement against standard PSG. We conclude that the recording of sleep with a biometric shirt system and the quantification of vigilance states using the HXC algorithm described here constitutes an optimal balance between portability and performance that deserves to be further investigated.

## ACKNOWLEDGEMENTS

The authors thank Ms Élyse Chevrier and Ms Marjolaine Chicoine from the Sleep Laboratory at the Hôpital Rivière-des-Prairies for their participation in the project. The study was partly supported by grants from the Fondation *Les petits trésors de l'Hôpital Rivière-des-Prairies* and the Natural Sciences and Engineering Research Council of Canada (NSERCC) to R. G. Ms Joëlle Pion-Massicotte received a scholarship from the Industrial Innovation Scholarships Program of the NSERCC and the *Fonds de recherche du Québec – Nature et Technologies*.

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## CONFLICT OF INTEREST

J. P. M.: was a trainee in engineering at Carré Technologies Inc., manufacturer of the Hexoskin biometric shirt, when the research was performed. Part of her scholarship was paid by Carré Technologies Inc. Now employee of Carré Technologies Inc. P. S.: none. J. F. R.: Chief Technology Officer and co-founder, Carré Technologies Inc., manufacturer of the Hexoskin biometric shirt. R. G.: none.

## 5 | AUTHOR CONTRIBUTORSHIP

J. P. M.: performed the literature survey, selected and developed the signal processing algorithms, recruited the participants, performed the statistical analyses, wrote the manuscript. R. G.: designed the study, screened the participants upon recruitment for the exclusions criteria using home-made questionnaires and a semi-structured interview, supervised sleep recordings and analyses, wrote the manuscript. P. S.: supervised the selection and development of the signal processing algorithms, supervised the statistical evaluation of the results, wrote the manuscript. J. F. R.: supervised the signal preprocessing software.

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## SUPPORTING INFORMATION

Additional Supporting Information may be found online in the supporting information tab for this article.

**How to cite this article:** Pion-Massicotte J, Godbout R, Savard P, Roy J-F. Development and validation of an algorithm for the study of sleep using a biometric shirt in young healthy adults. *J Sleep Res*. 2019;28:e12667. https://doi.org/10.1111/jsr.12667

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